SE311 SPRING 2025-2026 / 25-03-2026 & 26-03-2026
Lab Work 05 – Abstract Factory Pattern

In this lab, we will implement a **Hardware Product System** for different kinds of brands (**Logitech**, **Razer**). Each brand has its own **mouse**, **keyboard** products.

Look at your **AbstractFactory.java** code structure to understand the key idea behind the pattern and map the participants in here with the existing ones in the code.

Each product has its own technical specifications for demonstrating its performance on the system, which are stored as attributes in the concrete classes (e.g., **DPI** for mouse and **hz** for keyboards). Also, they have an **`executeFunction()`** method which will basically print a meaningful action, such as: *"LogitechMouse clicked with 12000 DPI sensitivity."*.

These products are created by subtypes of **HardwareFactory** for each brand type with some specific object creational methods.

Implement a class called **`HardwareSetupManager`** for managing your **`HardwareProducts`**. This class will also have a method called **`assembleHardwareSetup(HardwareFactory factory)`**. In that method, you will create each **`HardwareProduct`** for that factory type and add them into some list of products. Also, implement a method called **`runDiagnosticTest()`** to simply call each product's **`executeFunction()`** method in that list.

Whenever you are finished with these tasks, you will add another type of product called **Headphone** (related attribute for headphones is **frequency**) for all the types of abstract factories.

Also, implement another type of abstract factory for **`SteelSeries`** and make the necessary implementations.

Additionally, update your abstract factory. Instead of having different create methods for every product; it should have **one method accepting a parameter**, which indicates the type of product needs to be created.

Finally, make every factory as **SINGLETON** and in your main method create a **`HardwareSetupManager`** object and execute assembleHardwareSetup method for every factory type.